

Fruit Image Classification using Convolutional Neural Network (CNN)

This project focuses on image classification using a Convolutional Neural Network (CNN). The main objective is to classify fruit images into five different categories using a deep learning model built from scratch. TensorFlow and Keras libraries were used to design, train, and evaluate the CNN model.

Dataset Description

The dataset contains images of fruits arranged into training and testing folders. Five classes were selected for the project:

Selected Classes
Apple Red
Banana
Citrus Orange
Mango
Strawberry

CNN Architecture

The CNN model was implemented from scratch without using pretrained models. The architecture contains the following layers:

- Convolutional Layers (Conv2D)
- MaxPooling Layers
- Flatten Layer
- Dense Fully Connected Layer
- Dropout Layer
- Softmax Output Layer

The convolution layers help extract important image features such as edges, colors, and shapes. Pooling layers reduce image dimensions and improve computational efficiency.

Training Process

The images were resized to 128x128 pixels and normalized using ImageDataGenerator. The CNN model was trained using the Adam optimizer and categorical crossentropy loss function. Training was performed for multiple epochs using the training dataset, while the testing dataset was used for validation and evaluation.

Evaluation Metrics

Different evaluation metrics were used to analyze the performance of the model:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Project Results

The CNN model achieved approximately 98% accuracy during model evaluation. The training and validation accuracy graphs indicated stable model learning. Additional classification metrics such as precision, recall, and F1-score were also generated using the classification report and confusion matrix.

Classification Metrics

Epoch 7/10
134/134 155s 1s/step - accuracy: 0.9972 - loss: 0.0101 - val_accuracy: 0.9625 - val_loss: 0.3622

Epoch 8/10
134/134 159s 1s/step - accuracy: 0.9921 - loss: 0.0222 - val_accuracy: 0.9972 - val_loss: 0.0130

Epoch 9/10
134/134 160s 1s/step - accuracy: 0.9930 - loss: 0.0234 - val_accuracy: 0.9860 - val_loss: 0.0421

Epoch 10/10
134/134 153s 1s/step - accuracy: 0.9998 - loss: 0.0014 - val_accuracy: 0.9866 - val_loss: 0.0481

```
# Evaluate Model
test_loss, test_accuracy = model.evaluate(test_data)

print("Test Accuracy:", test_accuracy)
```

56/56 17s 299ms/step - accuracy: 0.9866 - loss: 0.0481
Test Accuracy: 0.9865696430206299

Classification Report:

	precision	recall	f1-score	support
Apple Red	0.26	0.27	0.26	500
Banana	0.19	0.19	0.19	327
Citrus Orange	0.19	0.19	0.19	278
Mango	0.17	0.15	0.16	289
Strawberry	0.26	0.26	0.26	393
accuracy			0.22	1787
macro avg	0.21	0.21	0.21	1787
weighted avg	0.22	0.22	0.22	1787

Confusion Matrix:

```
[[134 106 79 73 108]
 [111 62 49 52 53]
 [ 89 40 52 43 54]
 [ 78 53 38 44 76]
 [112 66 60 54 101]]
```

Conclusion

This project successfully demonstrated image classification using a simple CNN architecture. The model was able to classify fruit images effectively using deep learning techniques. The project improved understanding of convolutional neural networks, image preprocessing, model training, evaluation metrics, and classification performance analysis.